



ECE Research Seminar

April 17, 2026, Friday, 12:00 – 1:00 pm, Simrall 104

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A Brain-Inspired Autonomous Robotics Navigation: The Goal-Directed Cognitive Map

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Abstract: Autonomous systems and robots have traditionally relied on modular pipelines that separately and systemically address perception, mapping, localization, global planning, local reactive navigation, sensor configuration, and local control. While these approaches have enabled significant advances, they often struggle in uncertain, complex, dynamic, unforeseen, and partially observable environments. In contrast, biological systems demonstrate remarkable robustness, efficiency, and adaptability when navigating under uncertainty. This seminar explores brain-inspired intelligence including navigation and mapping through the lens of neuroscience, robotics, machine learning, and recent advances in whole-brain mapping and emulation. Recent breakthroughs in reconstructing the complete neural wiring of the fruit fly brain provide a compelling new opportunity to study navigation and mapping as an integrated cognitive process rather than a collection of independent algorithms. Inspired by biological mechanisms such as cognitive maps, place cells, grid cells, and head-direction cells, this talk presents the Goal-Directed Cognitive Map (GDCM), a biologically grounded navigation and mapping framework that unifies global path planning, local obstacle avoidance, mapping, and sensor-driven perception. By integrating brain-inspired representations with modern planning and control methods, GDCM enables robust, goal-directed navigation and mapping in complex, dynamic, and uncertain environments. The presentation highlights algorithm development, simulation results, system design insights, and future directions toward more cognitively grounded, and potentially emulated, navigation systems for autonomous robots based on the mammalian brain.



Matt Hicks is a Ph.D. student in the Department of Electrical and Computer Engineering at Mississippi State University, working under the supervision of Dr. Chaomin Luo as part of the Robotics and Intelligent Systems research group. He received his Bachelor of Science and Master of Science degrees in Electrical Engineering from the University of Washington in 2015 and 2018, respectively. His research interests include autonomous robotics navigation, brain-inspired algorithms, cognitive maps, neural representations of space, and biologically motivated path planning. He has authored and co-authored journal and multiple conference publications in IEEE venues on goal-directed

cognitive mapping and bio-inspired autonomous navigation systems such as IEEE TCDS, IEEE ROBIO, IEEE CARS, and IEEE CEC. Mr. Hicks is a full-time engineer at the Allen Institute in Seattle, a nonprofit bioscience and neuroscience research organization.

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